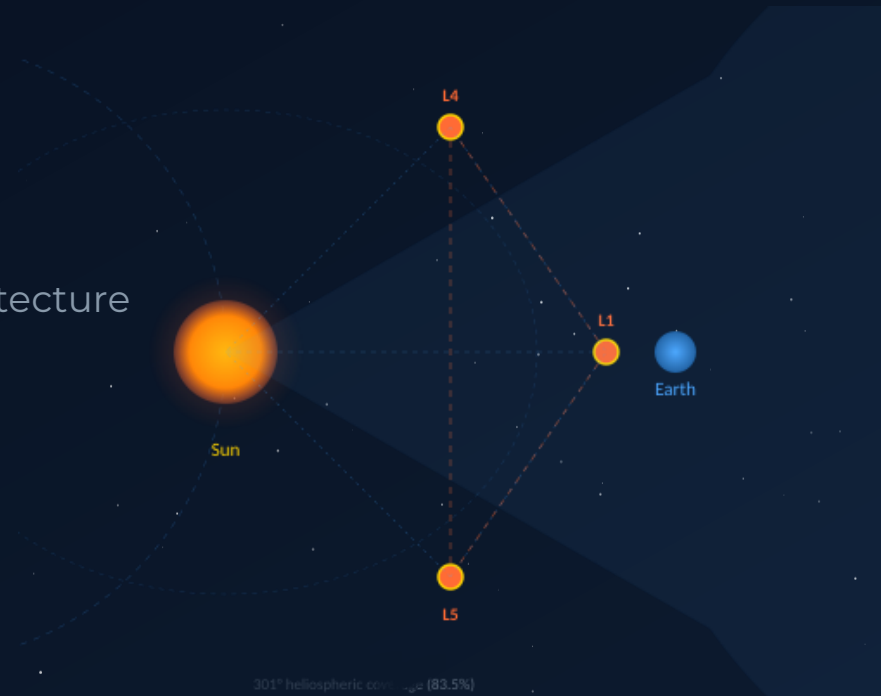


# HELIOS

Autonomous Solar Radiation Warning Architecture  
for Human Deep-Space Operations

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Paweł Micał | Space AI 2026



*"Protecting Human Life Beyond Earth's Magnetosphere"*

# ! THE PROBLEM

*"August 4, 1972 — between Apollo 16 and 17 — a solar storm delivered a lethal dose to deep space. Had astronauts been on the Moon, they would have received 4,000 mSv. The NASA limit is 250 mSv."*

## 15-60 min Warning

Current systems detect CME magnetic structure only when it physically passes L1. Insufficient for crew protective action.

## Earth-Centric Design

All existing systems require Earth-based manual analysis. No autonomous onboard crew warning capability exists.

## Limited Coverage

Single-point L1 observation. Less than 50% solar surface monitored. Major geometric blind spots.

# 985 mSv

Bastille Day dose  
(394% NASA limit)

# 40 sats

Starlink loss  
2022 storm

# 58,000

Satellites by 2030  
(10× current)

# \$42B

Potential daily losses  
from severe storm

# THE SOLUTION — HELIOS



## Distributed Architecture

L1 (validation) + L4/L5 (stereoscopic observation). Three-node constellation spanning 301° of the heliosphere.



## Autonomous AI Onboard

Neural network infers CME magnetic structure (Bz). No Earth dependency. TMR fault-tolerant voting.



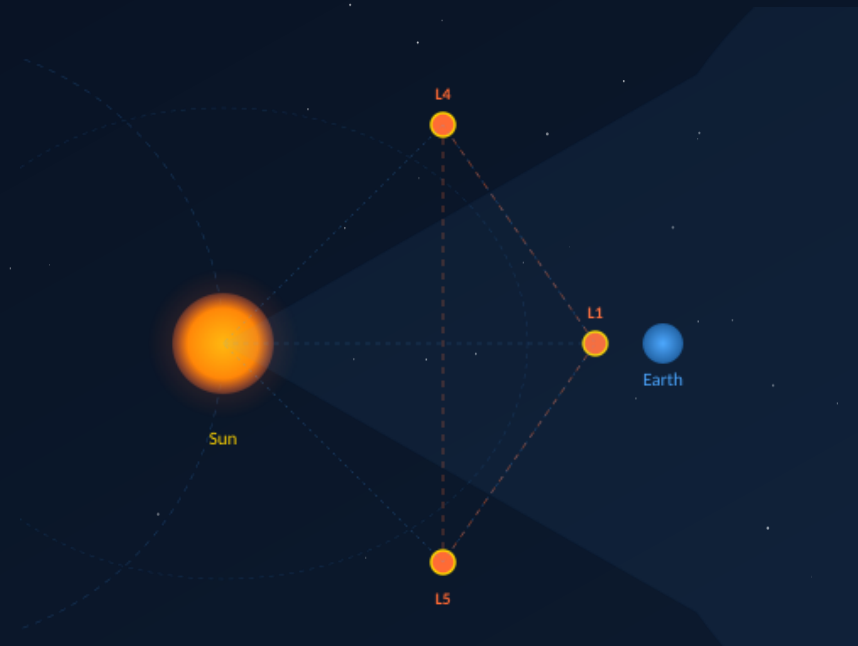
## 16-50 Hour Warning

Hours of advance notice vs. 15-60 minutes today. Speed-dependent: 50h for 800 km/s, 16h for 2500 km/s CME.



## Crew-Centric Design

Dose-based warnings in NASA-standard units (mSv). Severity classification: Low / Moderate / High / Extreme.



# 83.5%

Heliospheric coverage  
3.5-10.9h earlier detection  
vs single-point L1

# HOW IT WORKS

## 01

### DETECT

0.2–7.2h

Multi-perspective coronagraph imaging across L1/L4/L5. Running-difference algorithm triggers pipeline.

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**83.3% severity accuracy**

## 02

### TRIANGULATE

1–16h

Stereoscopic 3D reconstruction via Monte Carlo triangulation. Overdetermined 3-satellite geometry.

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**1.1M km resolution**

## 03

### INFER Bz

Onboard

Ensemble neural network (3 seeds) predicts magnetic field orientation. Heteroscedastic uncertainty.

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**6.3 nT MAE**

## 04

### WARN CREW

1–10 min

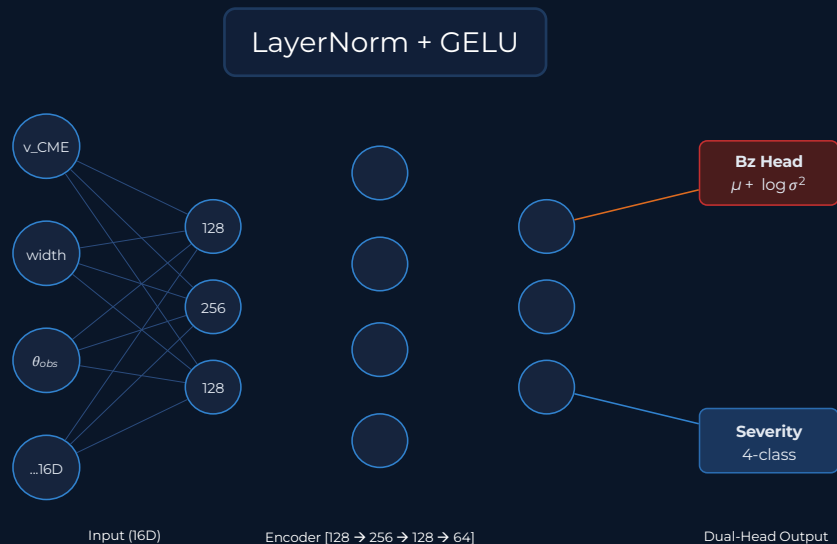
TMR voting consensus. Physics-based dosimetry calculation. Autonomous severity classification.

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**83.3% accuracy**

*First system to predict CME magnetic structure BEFORE it reaches L1*

# AI/ML ARCHITECTURE



## Feature Vector

16-dimensional: v\_CME, width,  $\theta_{obs}$ , parallax, triangulation quality, ...

## Encoder

[16→128→256→128→64] with LayerNorm + GELU activation

## Bz Head

Heteroscedastic regression: outputs  $\mu(Bz) + \log \sigma^2$  for uncertainty

## Severity Head

4-class classifier: Low / Moderate / High / Extreme

## Loss Function

Multi-task:  $\alpha=0.5$  (Gaussian NLL) +  $\beta=0.5$  (Cross-Entropy)

## Ensemble

3-seed training (42, 123, 456). Mean aggregation. 2.7 nT spread

## Training Pipeline

10,000 synthetic events (physics-based flux rope models) + 19 historical CMEs (12 train, 6 test, 1 unseen showcase) with 50× oversampling = 10,600 training samples

*Innovation: Physics-informed architecture · Uncertainty quantification · Fault-tolerant consensus · No data leakage*

# ✓ VALIDATED PERFORMANCE

## Bastille Day 2000 — Showcase Event

Predicted Bz: **-54.9 nT**

Actual Bz: **-60.0 nT**

**Error: 5.2 nT (8.6%)**

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Predicted Dose: 985 mSv (394% NASA 250 mSv SPE limit)

Classification: EXTREME ✓ (correct)

## Aggregate Validation Metrics

Bastille Day Confidence **91.6%**

Improvement vs Baseline **36%**

Bz MAE (aggregate) **6.3 nT**

Bz MAE (Bastille Day) **5.2 nT**

Severity Accuracy **83.3%**

Adjacent-or-Correct **100%**

## Validation Methodology

- ▶ 6 independent historical events (test set) — zero data leakage between train/test splits
- ▶ 3-seed ensemble with mean aggregation — inter-model spread: 2.7 nT (robust consensus)
- ▶ Physics-based dosimetry formula calibrated on 8 historical SPEs (1972–2017)
- ▶ Full pipeline: open-source GitHub repository — every metric independently reproducible

# ORBITAL MECHANICS

## L1 Halo Orbit Parameters

|                           |                           |
|---------------------------|---------------------------|
| Ax amplitude              | 220,908 km                |
| Ay amplitude              | 705,322 km                |
| Az amplitude              | 249,993 km                |
| Period                    | 177.74 days               |
| SEZ min. separation       | 9.51° (compliant)         |
| Stability index ( $\nu$ ) | 804 (active control req.) |
| HOI $\Delta V$            | ~50 m/s                   |
| Station-keeping           | 4.0 m/s/year              |
| 12-year propellant        | 17.2 kg                   |

## Insertion Innovation

Large-amplitude L1 Halo Orbit enables stable manifold insertion — dramatically reducing HOI delta-V vs direct insertion. L4/L5 nodes use Tadpole Orbit Insertion (TOI).

**HELIOS L1** ~50 m/s

**HELIOS L4/L5** ~50 m/s

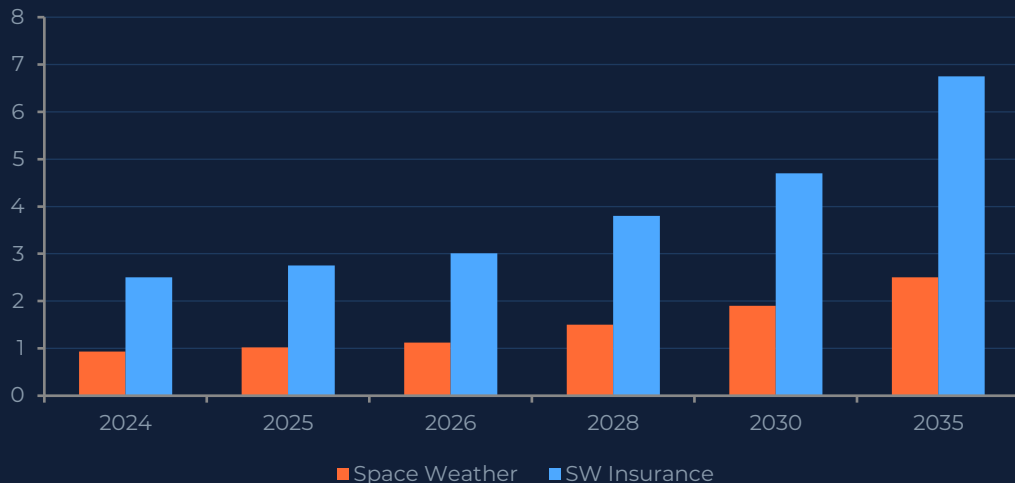
## Constellation Delta-V Budget

| Node | HOI/TOI | SK/yr   | Propellant |
|------|---------|---------|------------|
| L1   | ~50 m/s | 4.0 m/s | 17.2 kg    |
| L4   | ~50 m/s | 1.5 m/s | 14.3 kg    |
| L5   | ~50 m/s | 1.5 m/s | 14.3 kg    |

*Validated: Differential correction • Monodromy matrix stability*

# MARKET OPPORTUNITY

Market Growth (\$B)



## WHY NOW?

- ▶ **Artemis II** — 2026: First crewed lunar flight since Apollo
- ▶ **SOLAR-1** operational mid-2026 — still single-point, still 15-60 min warning
- ▶ **58,000** satellites projected by 2030
- ▶ **Munich Re** (Jan 2026): New space weather insurance products launched
- ▶ **JHU-APL**: "2026 will see greater focus on space weather prediction"
- ▶ Focus on **space datacenters** (Starcloud, SpaceX)

**\$2.5B**

Space Weather  
Market 2035

**\$6.75B**

SW Insurance  
Market 2035

**\$374B**

Space Infrastructure  
Market 2034

**58,000**

Satellites  
by 2030



# TARGET CUSTOMER SEGMENTS

## TIER 1 — Crew Safety (Life-Critical)

### Moon Bases

SpaceX HLS, NASA Artemis, ESA

\$100K–\$1M/mission

### Private Stations

Axiom Space, Vast, Blue Origin

\$500K–\$5M/yr

### Space Tourism

SpaceX Crew Dragon, Axiom Space, Vast

\$10K–\$100K/flight

## TIER 2 — Asset Protection

### Mega-Constellations

Starlink, Kuiper, OneWeb

\$500K–\$5M/yr

### Space Datacenters

SpaceX, Starcloud

\$1M+/yr

### Space Power Grids

Starcatcher, SSP operators

\$1M+/yr

## TIER 3 — Terrestrial & Science

### Aviation

Airlines, Boeing, Airbus

\$50K–\$500K/yr

### Scientific Data

NASA, ESA, JAXA, universities

\$100K–\$1M/yr

### Insurance

Munich Re, Tokio Marine,  
Swiss Re

Revenue share

## Value Chain Logic:

Artemis crew needs → Protection mandate → Station operators adopt → Insurance requires → Constellation operators follow → Market standard

# COMPETITIVE LANDSCAPE

| Capability        | HELIOS               | Mission Space       | ESA Vigil            | NOAA SOLAR-I        |
|-------------------|----------------------|---------------------|----------------------|---------------------|
| Architecture      | L1+L4+L5 distributed | LEO constellation   | Single L5            | Single L1           |
| Bz Prediction     | ✓ < 6 nT MAE         | ✗ Not possible      | ✗ No (In-situ at L5) | ✗ No (In-situ only) |
| Onboard AI        | ✓ TMR voting         | ✗ None              | ✗ None               | ✗ None              |
| Crew Warnings     | ✓ Autonomous         | ✗ None              | ✗ None               | ✗ None              |
| Warning Time      | 16-50 hours          | "96h" (unvalidated) | 4-5 days (rotation)  | 15-60 minutes       |
| Peer-Reviewed     | Planned (Q2 2026)    | ✗ No validation     | ✓ ESAPhase (C/D)     | ✓ NASA              |
| Hardware in Space | ✗ None               | ✓ ZOHAR-I (LEO)     | ✗ None (2031 launch) | ✓ Operational       |

## HELIOS Unique Positioning

- ▶ Only system designed for autonomous deep-space crew protection (vs Earth-centric approaches)
- ▶ Distributed L1+L4+L5 observation — new data from L4/L5 currently unavailable globally
- ▶ Early magnetic structure (Bz) inference — hours before L1 in-situ measurement (physically impossible for competitors)

# \$ BUSINESS MODEL & REVENUE

## Government Contracts

35%

NASA/ESA partnerships, DoD/Space Force

\$2-10M/yr

## Data Subscription

40%

Tiered: Basic alerts → Full API → Enterprise integration

\$5-20M/yr

## Mission Integration

15%

Spacecraft/habitat integration, per-mission licensing

\$2-10M/yr

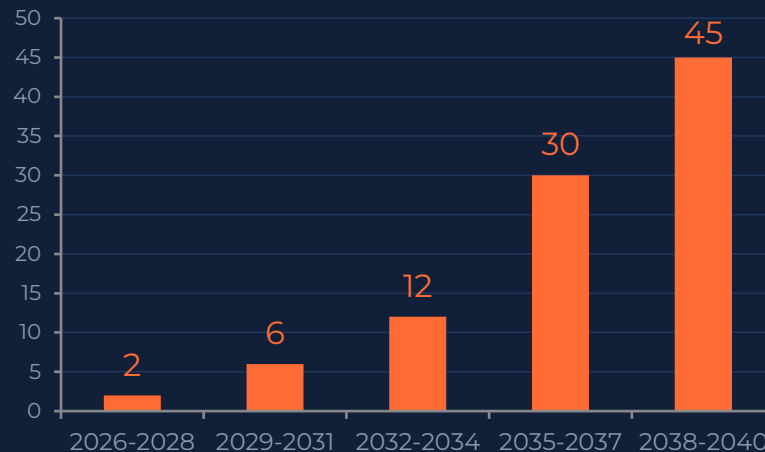
## Insurance Intel

10%

Risk data for Munich Re, Swiss Re, Tokio Marine HCC

\$1-5M/yr

Revenue Projection (\$M)



## Cost Structure (Estimates)

Satellite constellation (3 nodes) **\$50-150M** (phased over 3 years)

Launch (rideshare per node) **\$5-15M** (SpaceX Falcon 9)

Ground operations/year **\$2-5M** (3-5 global stations)

Team (20-40 people)/year **\$3-7M** (engineering + BD)



# GO-TO-MARKET STRATEGY

## PHASE 1

2026–2027

### Validation & Founding

- ▶ Peer-reviewed publication
- ▶ ESA BIC Incubation application
- ▶ Team building
- ▶ Seed round: \$2M–\$5M

## PHASE 2

2028–2029

### Satellite PDR & Partnerships

- ▶ Preliminary design review
- ▶ ESA Vigil data partnership
- ▶ Customer needs evaluation
- ▶ Series A: \$20M–\$50M

## PHASE 3

2030+

### Satellite CDR & Build

- ▶ Critical design review
- ▶ Flight qualification
- ▶ First launch target ~2033
- ▶ Essential utility status

# DOCUMENTATION & TECHNICAL STACK



## Technical Whitepaper

Complete system architecture, validation and analysis

- ▶ Appendix A: Orbital Mechanics (CR3BP dynamics)
- ▶ Appendix B: AI/ML Pipeline (architecture, training, validation)



## GitHub Repository

Open-source · Fully reproducible · Documented

- ▶ `helios_code/` — Detection, triangulation, ensemble propagation
- ▶ `NeuralNetwork_ML/` — Model architecture, training, TMR voting
- ▶ `helios_orbits/` — CR3BP dynamics, differential correction, SRP
- ▶ `output/` — Trained model + validated results (JSON)

## Technology Stack

### AI/ML

Python · PyTorch  
NumPy · SciPy  
LLM verification

### Physics Simulation

MATLAB/GMAT  
CR3BP Dynamics  
Monte Carlo

### Orbital Mechanics

Differential Correction  
Monodromy Matrix  
Tsiolkovsky Eq.

### Validation

6 Historical CMEs  
3-Seed Ensemble  
No Data Leakage

## MVP Validation Pipeline

Input: Historical CME data → Feature engineering (16D vector) → Ensemble inference (3 seeds) → TMR consensus → Dose calculation → Severity classification → Output: Warning JSON

*`run_complete_mvp.py`: Gold-standard script producing all whitepaper metrics — independently reproducible by any reviewer*

# TEAM & MILESTONES

## FOUNDER

**Paweł Micał**

17-year old student

- ▶ Sole architect of the core system
- ▶ Conceptualized the physics-informed ML pipeline and fault-tolerant system design
- ▶ Validated MVP: 18-page whitepaper + 2 appendices + open-source code
- ▶ Expert feedback from Dr. Martin Lo (Genesis mission) on Halo Orbit Insertion values

## MILESTONES

- ✓ **MVP Validation**  
83.3% severity accuracy, 5.2 nT Bz error
- ✓ **Technical Whitepaper**  
18 pages + 2 appendices
- ✓ **Open-Source Repository**  
Full validation pipeline on GitHub
- ✓ **Orbital Analysis**  
L1 halo orbit,  $\Delta V$  budget optimized
- ✓ **Expert Consultation**  
Dr. Martin Lo (Genesis) — Predicted HOI confirmed
- **Q2 2026: Publication**  
Peer-reviewed journal submission
- **Q3 2026: ESA BIC Incubation**  
Phase I application (€100K)
- **Q2 2027: Seed Round**  
\$2M–\$5M for team + satellite PDR
- **Q1 2028: ESA Partnership**  
ESA Vigil: Early Data Pipeline Integration

# THE VISION

*"By 2035, every human beyond Earth's magnetosphere will have HELIOS protection."*

## THE ASK

**Seed Funding: \$2M – \$5M**

- 40%** Engineering — satellite preliminary design, ground segment
- 30%** Team — CTO, BD lead, 2-3 engineers
- 20%** Business Development — customer pilots, conferences
- 10%** Operations — legal, IP protection, regulatory

## KEY NUMBERS

- \$2.5B** Market by 2035
- 83.3%** Severity accuracy (validated)
- 16-50h** Warning lead time
- 5.2 nT** Bz prediction accuracy
- \$30M** Projected Year 10 revenue

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*"HELIOS isn't just a satellite constellation. It's the difference between life and death for the first Mars crew."*